

Graph Ratio Tables

Lesson 3-3

Name:

Date:

Class:

My Notes

Level 3 Enrichment



TODAY'S OBJECTIVES

Content Objective: I can graph the values from a ratio table as points on the coordinate plane.

Language Objective: I can explain my graph using the words coordinate plane, ordered pair, origin, and proportional.



See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.



Fill in each blank as we go. Use the Word Bank to help you.



WORD BANK — FILL EACH BLANK WITH THE BEST WORD

Coordinate plane

Ordered pair

Linear pattern

Proportional

Origin



Tap any word to see what it means and a picture.

Level 3: try the blanks from memory first, then check the bank.

1

The flat grid formed by a horizontal and a vertical number line is the

2

A set of two numbers (x, y) that names a point is an .

3

When graphed points form a straight line, they show a .

4

Two quantities that always keep the same ratio are .

5

The point (0, 0) where the two axes meet is the .

Write About the Math

Guided ESOL writing

Use the support level you need. Every level answers the same math question.

1. Understand the Question

EXPLAIN

After plotting (1,2), (2,4), (3,6)... what pattern do you notice, and how does it show the chocolate-sauce ratio?

Your job: Answer the question. Use math evidence. Explain how the evidence proves your answer.

Responde la pregunta. Usa evidencia matemática. Explica cómo la evidencia demuestra tu respuesta.

2. Plan Your Math Words

Check at least two words you will use.

☐ **Coordinate plane** — A grid with a line going across and a line going up to plot points.
Plano cartesiano — Una cuadrícula con una línea horizontal y una vertical para marcar puntos.

☐ **Ordered pair** — Two numbers (x, y) that tell where a point is on a grid.
Par ordenado — Dos números (x, y) que dicen dónde está un punto en una cuadrícula.

☐ **Linear pattern** — Points that make a straight line on a grid.
Patrón lineal — Puntos que forman una línea recta en una cuadrícula.

☐ **Proportional** — Two amounts that grow together at the same rate.
Proporcional — Dos cantidades que crecen juntas al mismo ritmo.

☐ **Origin** — The point $(0, 0)$ where the two grid lines cross.
Origen — El punto $(0, 0)$ donde se cruzan las dos líneas de la cuadrícula.

Say it first: Say your idea to a partner or quietly to yourself before you write.

Di tu idea a un compañero o en voz baja antes de escribir.

The points form a ____ through the ____.

Los puntos forman una ____ que pasa por el ____.

3. Build Your Explanation

Model the parts: This model shows the parts of an explanation. It does not answer your writing question.

Claim: I predict the points will line up because sundaes and sauce grow together.

Evidence: A strong answer chooses sundaes (x) and ounces of sauce (y), and predicts a straight line because each sundae always needs 2 more ounces.

Reasoning: This evidence connects the definition of coordinate plane to the mathematical claim.

Start

Most language support

Complete the frames. Use the word bank.

Completa las oraciones. Usa el banco de palabras.

- **The points form a ____ through the ____.**
Los puntos forman una ____ que pasa por el ____.
- **As the number of sundaes goes up by 1, the ounces go up by ____.**
Cuando el número de copas sube en 1, las onzas suben en ____.

Build

Some language support

Write two connected sentences. Use because, so, or but.

Escribe dos oraciones conectadas. Usa porque, entonces o pero.

- **My claim is ____.**
Mi afirmación es ____.
- **I know because ____.**
Lo sé porque ____.
- **So ____.**
Entonces ____.

Explain

Light language support

Write a claim, give math evidence, and explain your reasoning.

Escribe una afirmación, da evidencia matemática y explica tu razonamiento.

- **My claim is ____.**
Mi afirmación es ____.
- **My math evidence is ____.**
Mi evidencia matemática es ____.
- **This proves ____ because ____.**
Esto demuestra ____ porque ____.

4. Check Your Explanation

- ☐ I answered the question.
Respondí la pregunta.

- ☐ I used math evidence: numbers, an equation, a model, or a comparison.
Usé evidencia matemática: números, una ecuación, un modelo o una comparación.

- ☐ I explained how the evidence proves my answer.
Explicué cómo la evidencia demuestra mi respuesta.

- ☐ I used at least two math words.
Usé por lo menos dos palabras matemáticas.

- ☐ I reread complete sentences and punctuation.
Volví a leer mis oraciones completas y la puntuación.